

Osmania University

SUPPLEMENTARY EXAMINATION ? 2026

Engineering Thermodynamics ? Supplementary Examination, 2026

Subject Code: ME301 Semester: 3 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: First Law of Thermodynamics)

1. State the First Law of Thermodynamics for a closed system and a control volume. Derive the steady flow energy equation. [12 Marks]
2. Solve a problem related to First Law of Thermodynamics. Show all the steps clearly. [8 Marks]
3. Define First Law of Thermodynamics. Explain its significance in Engineering Thermodynamics with a suitable example. [12 Marks]

SECTION B (Unit 2: Second Law of Thermodynamics)

4. State the Kelvin-Planck and Clausius statements of the Second Law of Thermodynamics. Prove their equivalence. [12 Marks]
5. Define entropy. Derive the Clausius inequality and explain the principle of increase of entropy. [12 Marks]
6. Explain Second Law of Thermodynamics with a neat diagram/sketch and a suitable example. [8 Marks]

SECTION C (Unit 3: Entropy and Availability)

7. Explain availability and irreversibility of a system with a suitable example and derive the expression for availability. [12 Marks]
8. Explain Entropy and Availability with a neat diagram/sketch and a suitable example. [12 Marks]
9. Explain Entropy and Availability with a neat diagram/sketch and a suitable example. [12 Marks]

SECTION D (Unit 4: Pure Substances)

10. Draw the P-V and T-S diagrams of the Carnot cycle and derive its efficiency. Explain the steam table properties. [12 Marks]
11. Explain Pure Substances with a neat diagram/sketch and a suitable example. [12 Marks]
12. How is Pure Substances implemented? Give an algorithm / procedure and explain. [12 Marks]

SECTION E (Unit 5: Ideal Gases and Mixtures)

13. Solve the given problem on ideal gas processes (isothermal, adiabatic and polytropic) using the characteristic gas equation. [8 Marks]

14. Describe the working principle of Ideal Gases and Mixtures with necessary equations.
[12 Marks]

15. Describe the working principle of Ideal Gases and Mixtures with necessary equations.
[12 Marks]

*** End of Question Paper ***